DATA CENTRE / WASTE HEAT

Heat that would otherwise be lost



How a data centre could warm a public pool and a district-heating network

Engineering research • MATLAB / Simulink • 2025 scenario

23 OCT 2026

PROJECT GLOSSARY · 1/2

Six terms that make the rest readable

Servers produce heat; we study how it can be used safely.

Data centre
A facility with servers, networking, power and cooling.

IT load
Server electrical demand; this study assumes a 10 MW peak.

Waste heat
Heat that must be removed from operating equipment.

Coolant loop
Pipes, pumps and fluid used to transport heat.

Heat exchanger
Transfers heat between hydraulically isolated fluid streams.

District heating
A hot-water network distributing heat to buildings.

PROJECT GLOSSARY · 2/2

How it delivers heat and handles failures

Some terms describe efficiency; others describe IT safety.

Heat recovery
Reusing heat that would otherwise be rejected.

Heat pump (HP)
Uses electricity to raise heat to a usable temperature.

Thermal store (TES)
Stores heat to smooth supply and demand over time.

Backup boiler
Provides heat when recovery and storage cannot cover demand.

N+1 redundancy
Adds one standby unit to the number required in normal operation.

SCADA
Supervisory monitoring and control: values, alarms and trends.

Key distinction: 1 MW is power; 1 MWh is the energy delivered at 1 MW for one hour.

VARIABLE PASSPORT · 1/2

Six quantities used in diagrams and plots

MW denotes power, °C denotes temperature. MWh instead measures energy.

SYMBOL	UNIT	MEANING AND DATA ORIGIN
P_IT	MW	Server electrical input; load profile is modeled.
Q_avail	MW	Server heat available for recovery; modeled, not metered.
Q_pool	MW	Useful pool heating, including any heat-pump contribution.
Q_DH	MW	Useful heat-pump output delivered to district heating.
P_HP	MW	Electric input power to the heat-pump compressor.
T_pool	°C	Pool-water temperature; simulated, not measured.

For a heat pump: useful heat = extracted source heat + compressor electricity.

VARIABLE PASSPORT · 2/2

Annual energy and efficiency indicators

An absent ERF is deliberate: it cannot be reported as measured without meters.

SYMBOL	UNIT	MEANING AND DATA ORIGIN
E_IT	MWh	IT energy over a period; derived from the model load.
E_DC	MWh	Total facility energy, including cooling and other overhead; estimated.
E_src	MWh	Annual extracted server heat; this is not the regulated ERF.
COP	1	Useful heat output per unit of compressor electricity.
PUE	1	E_DC / E_IT; 1.36 is an assumption, not a site measurement.
ERF	1	Energy reuse factor at the DC boundary; site metering is missing.

Evidence: ASSUMED (10 MW, PUE) · SIMULATED (Simulink) · UNMEASURED (ERF, real COP).

THE PROBLEM

Servers do more than compute

Almost all electricity used by servers eventually becomes heat.

10 MW

assumed maximum IT load

Servers must be cooled 24/7

Even when the pool and the city need no heat.

The usual path — reject the heat outdoors

Our question: how much can be reused without compromising IT safety?

CONCEPT

One idea: give the heat a second destination

Not instead of IT cooling, but alongside it.

01

Data centre

Warms technical coolant

02

Heat recovery

Transfers heat across isolated loops

03

Useful heating

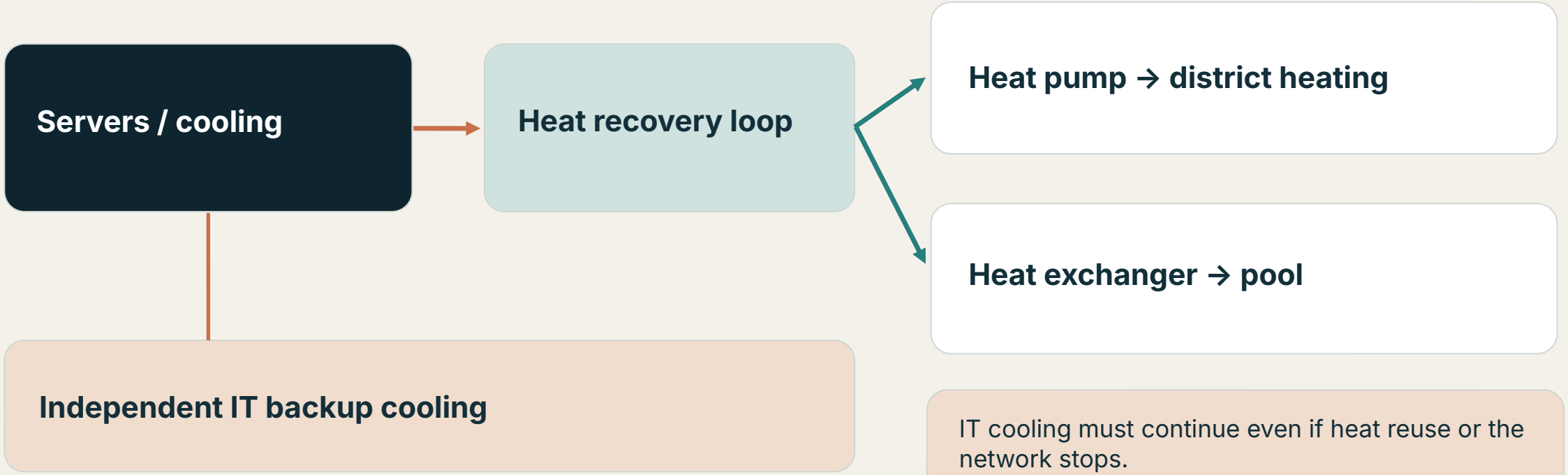
Pool + district network

Backup cooling remains independent of heat sales.

ARCHITECTURE

The system at a glance

Recover available server heat first; dispatch it to where it is needed.



ANNUAL BALANCE

What the simulated year shows

Server heat splits into recovered energy and energy rejected outdoors.

71.8

GWh of IT heat per year



38.3

GWh recovered

33.6

GWh rejected

This is a simulated energy balance, not a reportable measured ERF.

MODEL INPUTS

Where the numbers come from

Strength: reproducible calculations. Limitation: no operating-site telemetry.

Weather: external dataset

NASA POWER • Bratislava • 8,760 hours in 2025. Gridded reanalysis.

IT servers: assumed load

Load and recoverable fraction are model assumptions, not meter readings.

Physics: MATLAB / Simulink

Heat source, pool, heat pumps, storage and gas backup in 5-minute steps.

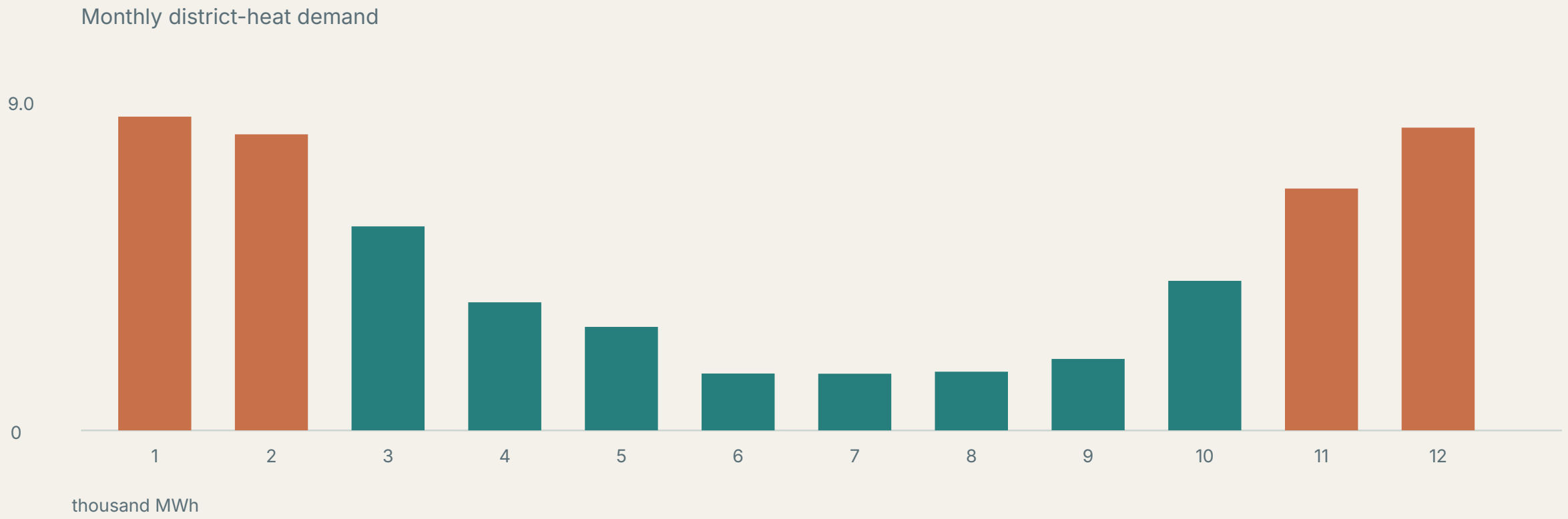
Verification: independent QA

Balance checks on 420,480 samples across four annual scenarios.

WEATHER AND DEMAND

Demand changes sharply with the season

District heat demand is calculated from Bratislava hourly outdoor temperature in 2025.



DISTRICT HEATING

How the heating demand is supplied

The heat pump supplies most of the heat; backup gas covers shortages and peaks.

56.8
GWh district demand

47.8
GWh useful HP heat

9.19
GWh backup gas heat



The storage tank shifts heat across time, so summed production flows include charge/discharge effects.

A PRACTICAL EXAMPLE

A swimming pool makes the benefit tangible

Assumed 25 × 12.5 × 1.6 m dimensions are a study scenario, not a universal pool standard.



25 × 12.5 × 1.6 m

500 m³
water volume

150 kW
heating capacity

250
MWh heat per year

27.62 °C
modeled peak

POOL PHYSICS

Where pool heat is gained and lost

The model includes water heat capacity, evaporation, convection, radiation and makeup water.

Temperature change

$$\Delta T = (Q_{in} - Q_{loss}) \cdot \Delta t / (\rho \cdot V \cdot c_p)$$

Qin: heat added

Qloss: heat lost

V: water volume

A pool cover reduces evaporation, and therefore heat demand.

TWO OPERATING MODES

35°C: direct. 25°C: via a heat pump

The available coolant temperature determines which equipment is practical.

35 °C

Direct transfer

Servers → Pool

No dedicated pool compressor if the exchanger achieves the required heat transfer.

25 °C

Heat pump

Source → Compressor → pool

Compressor electricity is needed; COP=4 is assumed, not manufacturer-certified.

ENERGY ACCOUNTING

A heat pump does not create energy

At 25°C the recovered source supplies heat; electricity drives the compressor.



= 250.1 MWh

useful pool heating over one year

DISTRICT HEAT PUMP

Why COP matters

COP measures useful heat delivered per unit of compressor electricity.

COP ≈ 4.90

calculated from a synthetic Carnot approximation

47.8 GWh

district useful heat

9.74 GWh compressor electricity

OEM certification is required at the actual 25–35°C source and 70°C sink conditions.

EQUIPMENT

The equipment we would need

This is a technical selection brief, not a purchase-ready bill of materials.

Pool: plate heat exchanger

150 kW; ≈25 m² at 35/32 ↔ 27.5/29.5°C with assumed heat transfer coefficient.

District network: modular HPs

Carrier or Oilon; module count requires OEM output/COP data for our temperatures.

Pumps and metering

Preliminary 43.2 m³/h primary and 64.8 m³/h pool-side flow.

Independent IT heat rejection

Full 10 MW rejection even with no heat customers, with N+1 resilience.

CATALOGUE CHECK

A 1.57 MW rating is not a site guarantee

A manufacturer rating refers to specific test temperatures, not every operating point.

1.57 MW

Carrier 61XWH-ZE17: published nominal point

6 + 1

nominal units for 9.42 MW + one spare

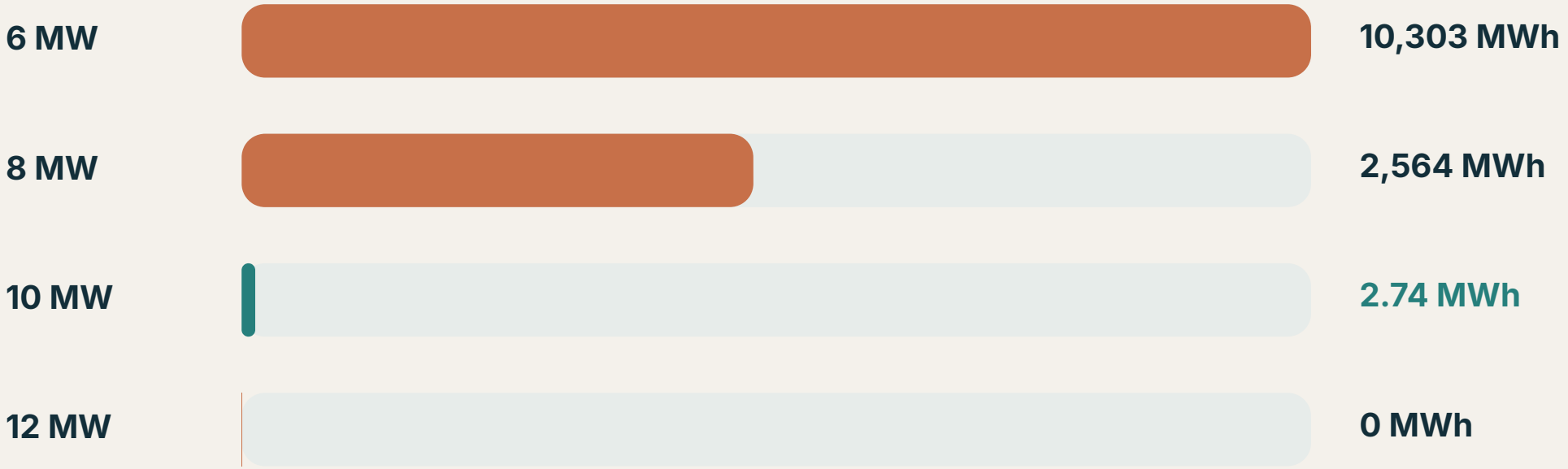
Published test point: 20/15°C source → 70/75°C sink.

For the intended 35/30°C → 50/70°C operation, OEM selection is still required. 6+1 is not a purchase decision.

CAPACITY CHOICE

Why the biggest plant is not automatically best

Compare peak hours, underused capacity and backup-gas substitution.



Extra annual backup-gas heat with constrained HP capacity

POOL ECONOMICS

The simpler route currently wins on cost

Against separate gas-boiler pool heating at Eurostat Slovakia 2025 benchmark prices.

€12,307

annual energy OPEX advantage: direct HX

€4,280

annual energy OPEX advantage: pool heat pump (COP=4 assumed)

Excludes installed CAPEX, maintenance, hall HVAC and actual contracted tariffs.

INVESTMENT DECISION

Installed cost can change the conclusion

Direct HX: 20-year NPV at 6% discount. These are budget scenarios, not supplier quotations.

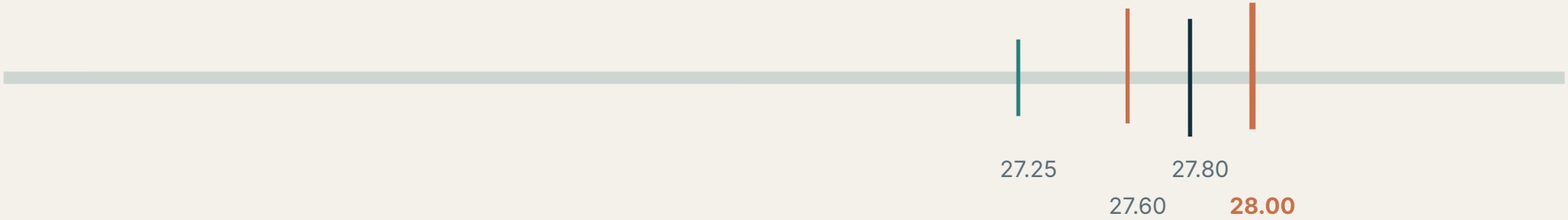
€45k	€75k	€110k
NPV +€71k	NPV +€34k	NPV €-9k

Under the model assumptions, break-even installed CAPEX is around €103k.

TEMPERATURE AND SAFETY

Comfort and a strict temperature boundary

The simulation deliberately stays below the published upper temperature limit.



27.62 °C

annual maximum in Simulink

Actual sensor errors, valve overshoot and effective legal text still require testing.

RESILIENCE

What if nobody accepts the heat?

Servers cannot depend on pool schedules or district heating demand.

10 MW

full-rated independent IT heat rejection required

1,727 m³/h

at a 5 K water temperature difference

N+1

survive failure of at least one unit

Thermal storage is not coolant flow: pumps require UPS and redundancy.

FAULT SCENARIOS

A useful model must reveal unsafe conditions

Independent cooling model: six one-hour tests at 5-second intervals, with a 10 m^3 buffer.

Spare starts in 30 sec	Within limit
Spare delayed 5 min	Overheat detected
Capacity derated in heat	Capacity shortage flagged
Coolant pump failure	No coolant flow — trip

Unsafe cases are successful hazard detections, not successful safe operation.

EVIDENCE AND STANDARDS

Verified model ≠ certified facility

We mapped standards but have not signed off compliance without full texts and tests.

34	56	36	14
defined KPIs	standards checkpoints	use cases	defined signals

22 / 36 passed software checks

14 remain blocked

PUE=1.36 is assumed. Reportable ERF and WUE require meters.

USE CASES

What different people gain

Visitors notice a warm pool; operators need a safe, measurable heat system.

Pool visitor

Comfortable water and potentially lower energy costs.

Pool operator

Heat forecast, water-temperature control and backup operation.

Data-centre operator

Recover heat without depending on external customers for IT cooling.

District network / city

An additional heat source, with reduced modeled backup-gas reliance.

NEXT STEPS

From research to a buildable design

The project needs staged evidence, not a leap from simulations to purchasing.

01 · Measure

IT loads, coolant temperatures/flows and facility energy

02 · Select

Heat pumps and HX from confirmed operating-point curves

03 · Engineer

Hydraulics, UPS, independent backup cooling and safeguards

04 · Validate

Fault testing, applicable standards and acceptance

Buying equipment before OEM ratings and safety testing would be premature.

What the study shows

- Server heat can be allocated among multiple useful destinations.
- At a 35°C source, direct pool heating appears simpler and cheaper than a separate HP.
- Heat reuse must never become the only IT cooling route.

A research conclusion, not an installation approval.

REFERENCE

How to interpret every number in this deck

All major numbers come from saved model results with explicit evidence limits.

CALCULATED

2025 model, 8,760 hours, 300-second Simulink, independent balance QA.

ASSUMED

10 MW IT, PUE=1.36, modeled COP, equipment budgets, 500 m^3 pool.

STILL UNVERIFIED

OEM duty/COP, site measurements, physical safety and licensed standards.

PRIMARY SOURCES

NASA POWER • Eurostat SK • Carrier / Eurovent • EU 2024/1364 • STN EN 15288.

Supporting files: MATLAB/Simulink, KPI CSV/JSON, 56 standards checkpoints, 36 use cases and QA.